

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class: TYIT

Sem: V

Roll No.: TYIT16

Date : 04/07/26

Course Name: IOT

Page no:

Practical Number:- 1A

AIM: A demonstrate traffic light using led and Arduino Uno on a simulation software.

STEPS:

1. On the tinkercad ,click on the create circuit .
2. Search arduino uno r3,breadboard,and leds and resistors.
3. Leds are placed in breadboard like red led in d3,yellow led in d13,and led in 24 respectively.
4. Resistors are placed parallel to led, one is connected to led and other one on GND(negative).
5. On arduino in digital side the wire is placed in 3.3v and it is connected to the positive of the breadboard and GND is connected to breadboard GND(negative).
6. At the digital side, 3 wire are pinned in 4,-3, and 2 and the other end is connected were led and resistor are pinned.
7. After connection and code we can start the simulation

CODE:

```
int red = 4;
int yellow = 3;
int green = 2;

void setup()
{
  pinMode(red,OUTPUT);
  pinMode(yellow,OUTPUT);
  pinMode(green,OUTPUT);
}

void loop()
{
  digitalWrite(red, HIGH);
  digitalWrite(yellow, LOW);
  digitalWrite(green, LOW);
  delay(500);
  digitalWrite(red, LOW);
  digitalWrite(yellow, HIGH);
  digitalWrite(green, LOW);
  delay(500);
  digitalWrite(red, LOW);
  digitalWrite(yellow, LOW);
  digitalWrite(green, HIGH);
  delay(500);
}
```

Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class: TYIT

Sem: V

Roll No.:TYIT16

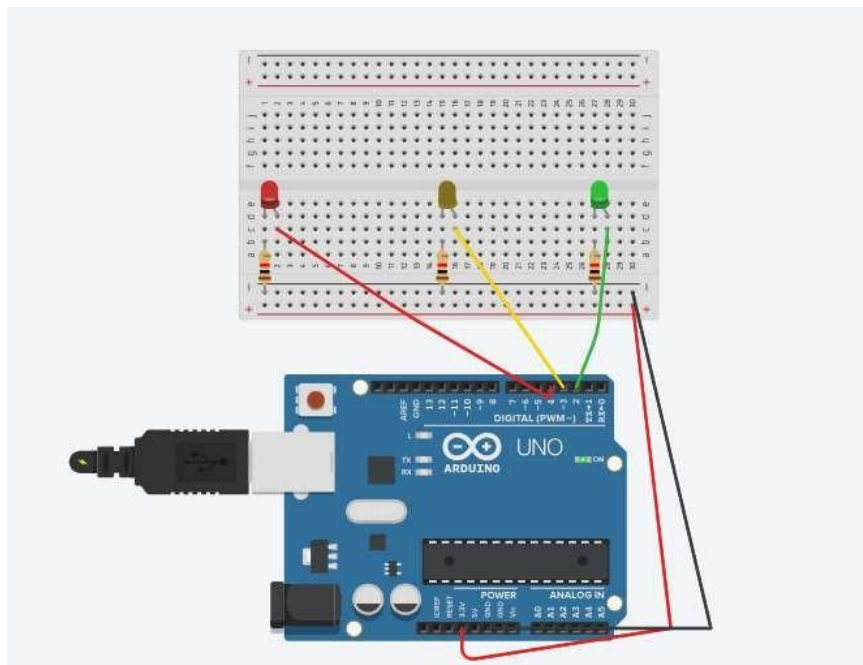
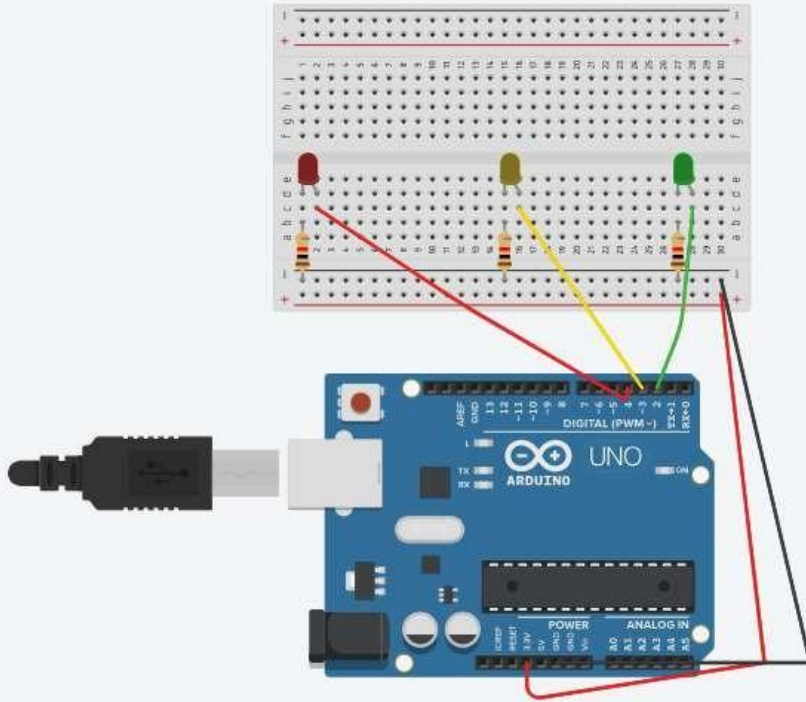
Date :04/07/26

Course Name: IOT

Page no:

Practical Number:- 1A

OUTPUT: CIRCUIT DIAGRAM



Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class:TYIT

Sem: V

Roll No.: TYIT16

Date :09/07/26

Course Name: Internet Of Things

Page no:

Practical Number:- 1b

AIM:Demonstrating the use of Pushbutton.

Steps:

- 1.On the Tinkercad,click on the create circuit.
- 2.Search Arduino uno r3,Breadboard,Leds,Resistors and Pushbutton.
- 3.Then place the LED,resistor and the pushbutton on the breadboard.
- 4.Then connect the LED with the Arduino (in d13) Digital side.
- 5.Also connect the Pushbutton with Arduino (in -3) Digital side.
- 6.Then the cathode(positive) is connected to the 3.3V on the Analog side.
- 7.The Anode(negative) is connected to the GND of the Analog side.
- 8.Also from the Breadboard connect the Cathode (positive) to the pushbutton.
- 9.After connection and code we can start the simulation.

Code:

```
int red = 13;
```

```
int buttonstate = 0;
```

```
int pushbutton= 3;
```

```
void setup()
```

```
{
```

```
  pinMode(pushbutton,INPUT);
```

```
  pinMode(red, OUTPUT);
```

```
}
```

```
void loop()
```

```
{
```

```
  buttonstate = digitalRead(3);
```

```
  if(buttonstate == HIGH)
```

```
  {
```

```
    digitalWrite(13, HIGH);
```

```
  }
```

```
  else{
```

```
    digitalWrite(13, LOW);
```

```
  }
```

```
  delay(10);
```

```
}
```

Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class:TYIT

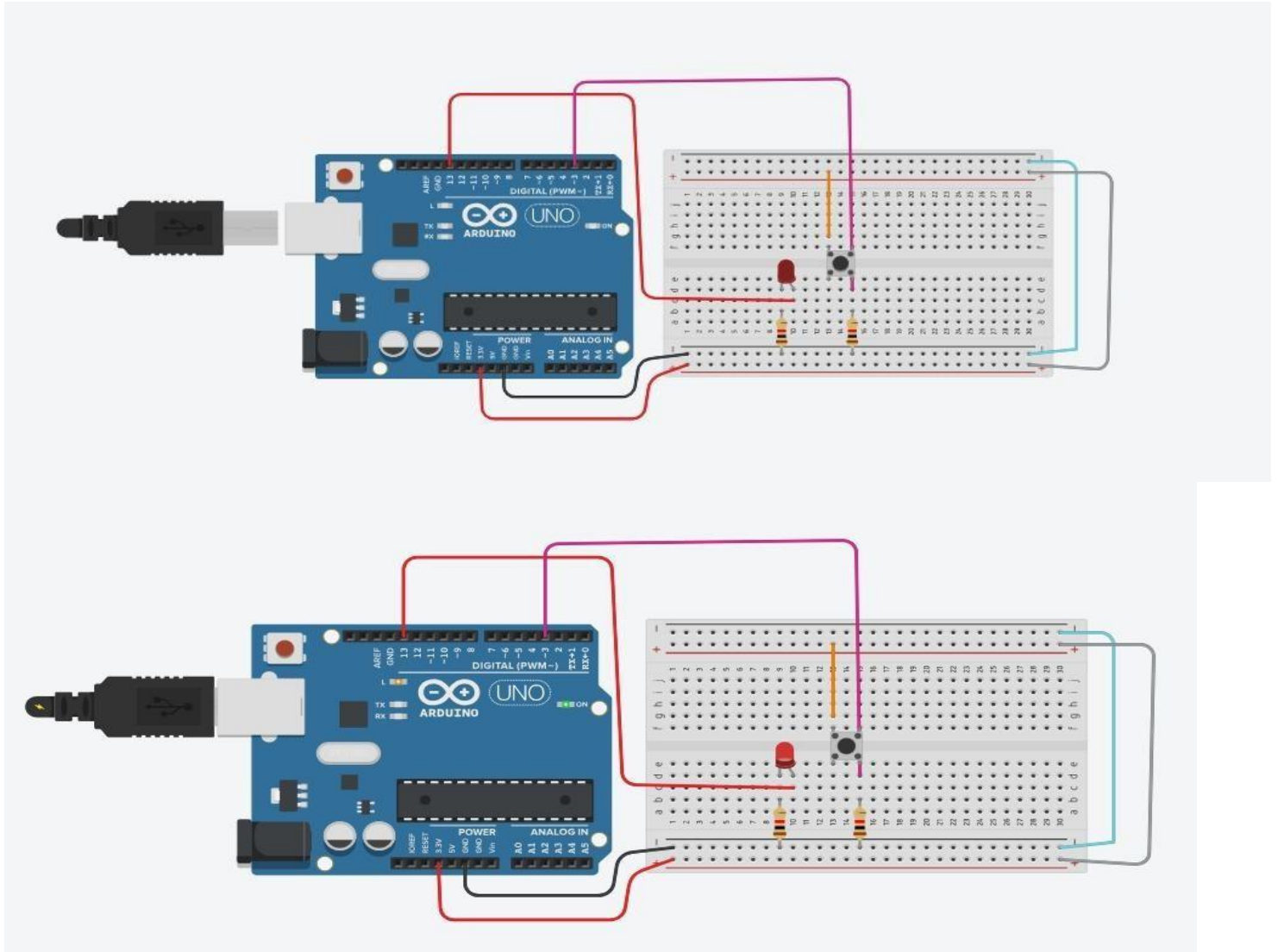
Sem: V

Roll No.: TYIT16

Date :09/07/26

Course Name: Internet Of Things
Practical Number:- 1b

Page no:



Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class:TYIT

Sem: V

Roll No.: TYIT16

Date :09/07/26

Course Name: Internet Of Things

Page no:

Practical Number:- 1c

AIM:Implement LED patterns by configuring multiple Push buttons.

Steps:

1. On the tinkercad ,click on the create circuit .
2. Search arduino uno r3, breadboard, 4 leds, 8 resistors, and 4 pushbutton.
3. Led are placed in d2,d5,d8 and d11 in the breadboard in which one wire is connected to digital 13 ,12,11,10 and 4 pushbutton are placed in center in which wire is connected to 6 ,5,4,3 and 2 .
4. On arduino in analog side the wire is placed in 3.3v and it is connected to the positive of the breadboard and GND is connected to breadboard GND(negative).
5. In breadboard , the positive and GND are connected to each other.
6. After connection and code we can start the simulation.

Code:

```
//----- LED Pins -----  
const int led1 = 13;  
const int led2 = 12;  
const int led3 = 11;  
const int led4 = 10;  
  
//----- Button Pins -----  
const int button1 = 6;  
const int button2 = 5;  
const int button3 = 4;  
const int button4 = 3;  
  
void setup() {  
  
    pinMode(led1, OUTPUT);  
    pinMode(led2, OUTPUT);  
    pinMode(led3, OUTPUT);  
    pinMode(led4, OUTPUT);  
  
    pinMode(button1, INPUT_PULLUP);  
    pinMode(button2, INPUT_PULLUP);  
    pinMode(button3, INPUT_PULLUP);  
    pinMode(button4, INPUT_PULLUP);  
}  
  
void loop() {  
  
    if (digitalRead(button1) == LOW) {
```

Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class:TYIT

Sem: V

Roll No.: TYIT16

Date :09/07/26

Course Name: Internet Of Things

Page no:

Practical Number:- 1c

```
    runningPattern();
}
else if (digitalRead(button2) == LOW) {
    alternatePattern();
}
else if (digitalRead(button3) == LOW) {
    knightRiderPattern();
}
else if (digitalRead(button4) == LOW) {
    UniquePattern();
}
else {
    allOff();
}
}
```

//-----

// Pattern 1 : Running LEDs

//-----

void runningPattern() {

```
    digitalWrite(led1, HIGH);
    delay(300);
    digitalWrite(led1, LOW);
```

```
    digitalWrite(led2, HIGH);
    delay(300);
    digitalWrite(led2, LOW);
```

```
    digitalWrite(led3, HIGH);
    delay(300);
    digitalWrite(led3, LOW);
```

```
    digitalWrite(led4, HIGH);
    delay(300);
    digitalWrite(led4, LOW);
```

}

//-----

// Pattern 2 : Alternate Blink

//-----

void alternatePattern() {

```
    digitalWrite(led1, HIGH);
    digitalWrite(led2, LOW);
```

Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class:TYIT

Sem: V

Roll No.: TYIT16

Date :09/07/26

Course Name: Internet Of Things

Page no:

Practical Number:- 1c

```
digitalWrite(led3, HIGH);  
digitalWrite(led4, LOW);  
delay(500);
```

```
digitalWrite(led1, LOW);  
digitalWrite(led2, HIGH);  
digitalWrite(led3, LOW);  
digitalWrite(led4, HIGH);  
delay(500);
```

```
}
```

```
//-----
```

```
// Pattern 3 : Knight Rider
```

```
//-----
```

```
void knightRiderPattern() {
```

```
    digitalWrite(led1, HIGH);  
    delay(500);  
    digitalWrite(led1, LOW);
```

```
    digitalWrite(led2, HIGH);  
    delay(500);  
    digitalWrite(led2, LOW);
```

```
    digitalWrite(led3, HIGH);  
    delay(500);  
    digitalWrite(led3, LOW);
```

```
    digitalWrite(led4, HIGH);  
    delay(500);  
    digitalWrite(led4, LOW);
```

```
    digitalWrite(led3, HIGH);  
    delay(500);  
    digitalWrite(led3, LOW);
```

```
    digitalWrite(led2, HIGH);  
    delay(500);  
    digitalWrite(led2, LOW);
```

```
    digitalWrite(led1, HIGH);  
    delay(500);  
    digitalWrite(led1, LOW);
```

Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class:TYIT

Sem: V

Roll No.: TYIT16

Date :09/07/26

Course Name: Internet Of Things

Page no:

Practical Number:- 1c

}

//-----

// Pattern 4 : Unique Pattern

//-----

```
void Bhumi () {  
    digitalWrite(led1, HIGH);  
    digitalWrite(led2, LOW);  
    digitalWrite(led3, LOW);  
    digitalWrite(led4, HIGH);  
    delay(3000);
```

```
    digitalWrite(led1, HIGH);  
    digitalWrite(led2, LOW);  
    digitalWrite(led3, LOW);  
    digitalWrite(led4, HIGH);  
    delay(3000);  
}
```

//-----

// Turn all LEDs OFF

//-----

```
void allOff() {
```

```
    digitalWrite(led1, LOW);  
    digitalWrite(led2, LOW);  
    digitalWrite(led3, LOW);
```

```
}
```

Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class:TYIT

Sem: V

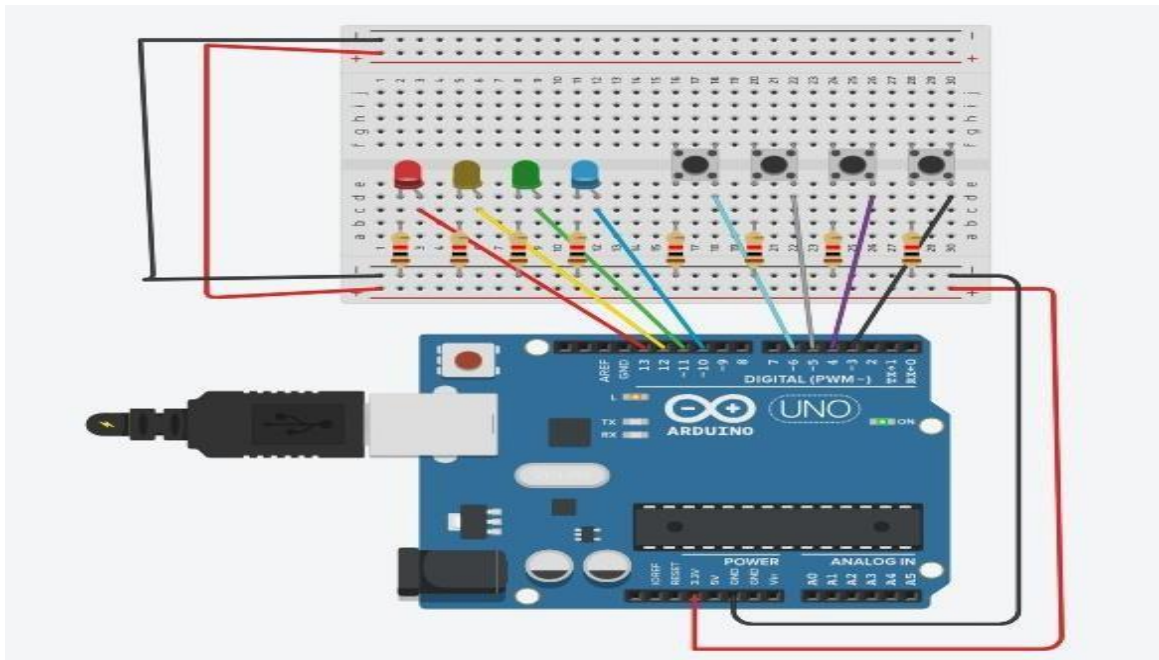
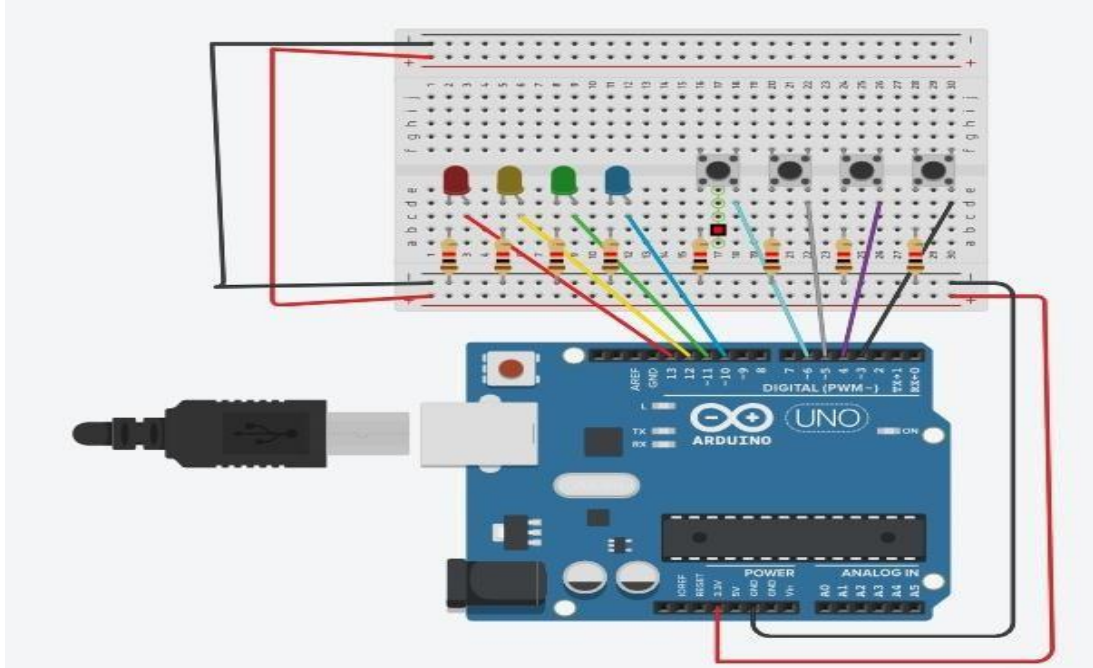
Roll No.: TYIT16

Date :09/07/26

Course Name: Internet Of Things
Practical Number:- 1c

Page no:

Output:



Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class: TYIT

Sem: V

Roll No.: TYIT16

Date :21/07/2026

Course Name:IOT

Page no:

Practical Number:-2

AIM(a):Interfacing 16x2 LCD Display with Arduino Board to Display text with scrolling effect.

Code:

```
#include <LiquidCrystal.h>
```

```
// RS, E, D4,D5, D6, D7
```

```
LiquidCrystal lcd(8, 7, 6, 5, 4, 3);
```

```
void setup() {  
  lcd.begin(16, 2);  
}
```

```
void loop() {
```

```
  for(int i = 1;i <= 60; i++) {
```

```
    //First row  
    lcd.setCursor(0, 0);  
    lcd.print("Class of TYIT:");
```

```
    //Create Roll Number  
    String roll = "TYIT" + String(i);
```

```
    //Add spaces before and after for scrolling effect  
    String scrollText= "      " + roll + "      ";
```

```
    //Scroll the roll number across the second row  
    for (int pos = 0; pos <= scrollText.length() - 16; pos++) {  
      lcd.setCursor(0, 1);    lcd.print(scrollText.substring(pos,  
pos + 16));    delay(50);  
    }
```

```
    delay(500);  
  }  
}
```

Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST PRACTICAL JOURNAL

Class: TYIT **Sem: V**

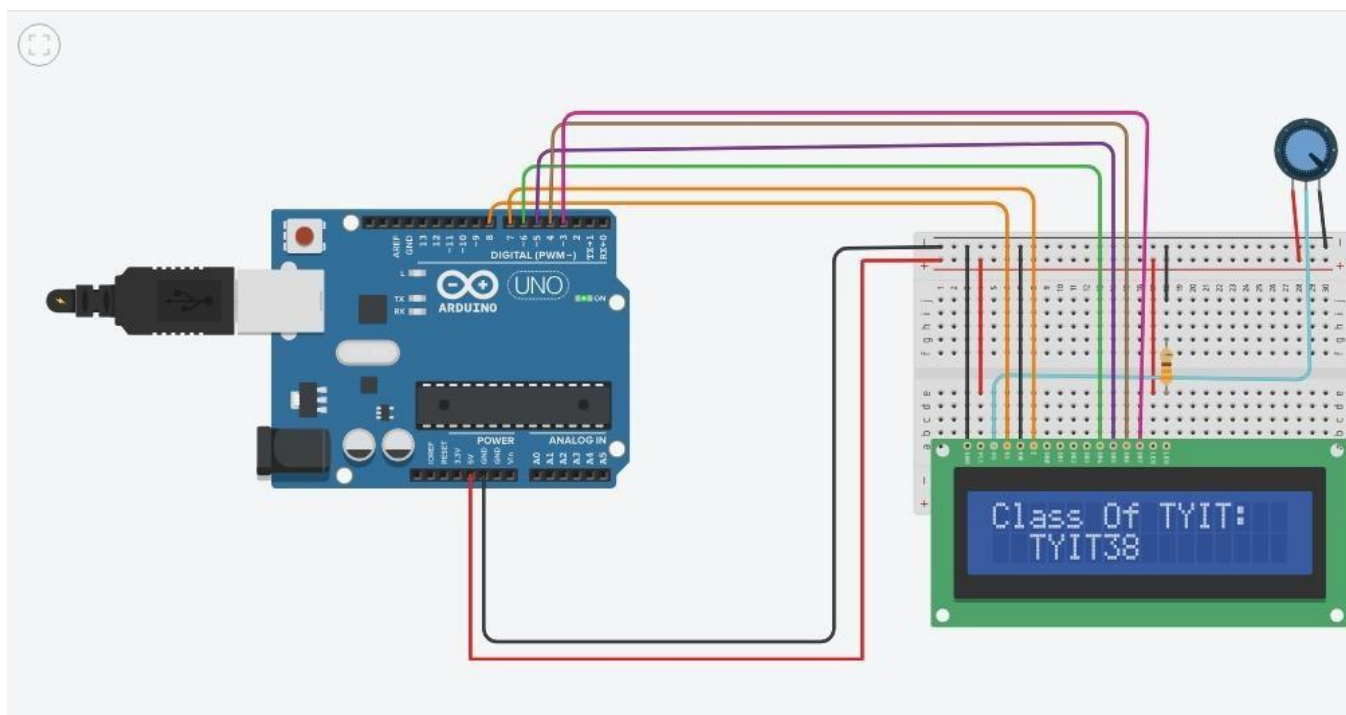
Class: TYIT Sem: V Roll No.: TYIT16

Class: TYIT Sem: V Roll No.: TYIT16 Date :21/07/2026

Page no:

Practical Number:-2

Output:



Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class: TYIT

Sem: V

Roll No.: TYIT16

Date :21/07/2026

Course Name:IOT

Page no:

Practical Number:-2

AIM(b): 2b.to Interfacing 16x2 LCD Display with Arduino Board Display a list of items.

CODE:

```
#include <LiquidCrystal.h>
```

```
// RS, E, D4, D5, D6, D7
```

```
LiquidCrystal lcd(8, 7, 6, 5, 4, 3);
```

```
String items[] = {
```

```
    "Monsoon",
```

```
    "Winter",
```

```
    "Autumn",
```

```
    "Spring",
```

```
    "Summer"
```

```
};
```

```
Void setup() {
```

```
    Lcd.begin(16, 2);
```

```
}
```

```
Void loop() {
```

```
    For (int I = 0; I < 5; i++) {
```

```
        Lcd.clear();
```

```
        // First Row
```

```
        Lcd.setCursor(0, 0);
```

```
        Lcd.print("List Seasons");
```

```
        // Second Row
```

```
        Lcd.setCursor(0, 1);
```

```
        Lcd.print(items[i]);
```

```
        Delay(2000);
```

```
    }
```

```
}
```

Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class: TYIT

Sem: V

Roll No.: TYIT16

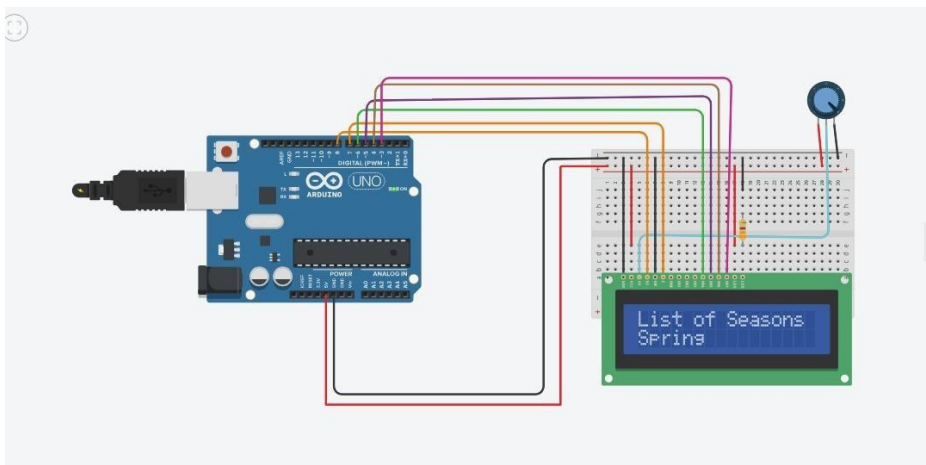
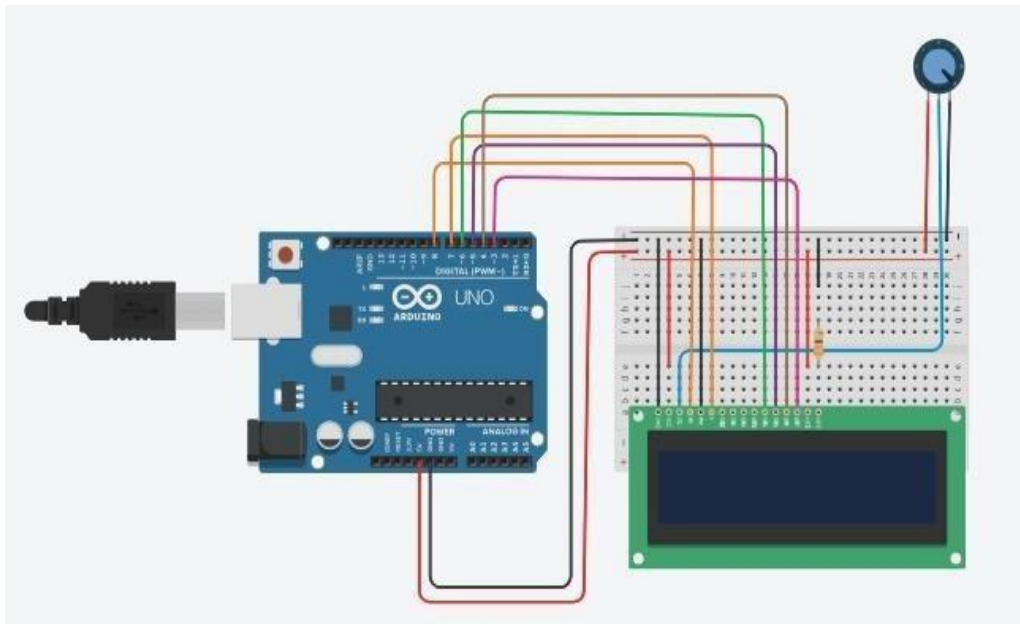
Date :21/07/2026

Course Name:IOT

Page no:

Practical Number:-2

OUTPUT:



Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class: TYIT(NEP)

Sem: V

Roll No.: TYIT16

Date:

Course Name: Internet Of Things

Page no:

Practical Number:-2b

Aim: Interface and lcd with Arduino Uno and display a list of items

Steps:

1. Open Tinkercad → Circuits → Create New Circuit.
2. Add an Arduino Uno, Breadboard, 16×2 LCD Display, 220 kΩ Potentiometer, and 220 Ω Resistor.
3. Connect the Arduino 5V pin to the + (red) power rail of the breadboard.
4. Connect the Arduino GND pin to the – (blue/black) power rail of the breadboard.
5. Place the 16×2 LCD on the breadboard.
6. Connect the LCD VSS pin to GND and LCD VDD pin to 5V.
7. Connect the LCD VO (Contrast) pin to the middle pin of the potentiometer.
8. Connect the two outer pins of the potentiometer to 5V and GND.
9. Connect the LCD RW pin to GND.
10. Connect the LCD RS pin to Arduino Digital Pin 8.
11. Connect the LCD E (Enable) pin to Arduino Digital Pin 7.
12. Connect the LCD D4, D5, D6, and D7 pins to Arduino Digital Pins 6, 5, 4, and 3 respectively.
13. Connect the LCD LED+ pin to 5V through the 220 Ω resistor.
14. Connect the LCD LED– pin to GND.
15. Check all the wiring connections carefully.
16. Upload the Arduino program.
17. Start the simulation and rotate the potentiometer until the text appears clearly on the LCD.

Code:

```
#include <LiquidCrystal.h>
```

```
// RS, E, D4, D5, D6, D7
LiquidCrystal lcd(8, 7, 6, 5, 4, 3);
```

```
// List of items
String items[] = {
  "Salt",
  "Cooking Oil",
  "Wheat Flour",
  "Eggs",
  "Cheese",
  "Tomatoes",
  "Potatoes",
  "Onions",
  "Toothpaste",
  "Detergent Powder"
};
```

```
int totalItems = sizeof(items) / sizeof(items[0]);
```

Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class: TYIT(NEP)

Sem: V

Roll No.: TYIT16

Date:

Course Name:Internet

Of Things

Page no:

Practical Number:-2b

```
void setup() {  
  lcd.begin(16, 2);  
}  
  
void loop() {  
  for (int i = 0; i < totalItems; i++) {  
  
    // First Row  
    lcd.setCursor(0, 0);  
    lcd.print("Item List:   "); // Extra spaces clear old text  
  
    // Current item  
    String scrollText = "          " + items[i] + "          ";  
  
    // Scroll item on second row  
    for (int pos = 0; pos <= scrollText.length() - 16; pos++) {  
      lcd.setCursor(0, 1);  
      lcd.print(scrollText.substring(pos, pos + 16));  
      delay(150);  
    }  
  
    delay(500);  
  }  
}
```

Circuit Diagram:

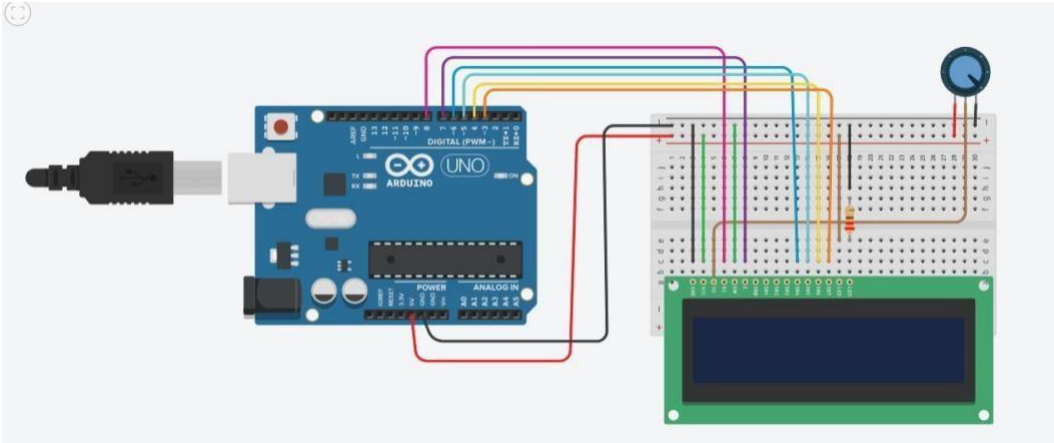
Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

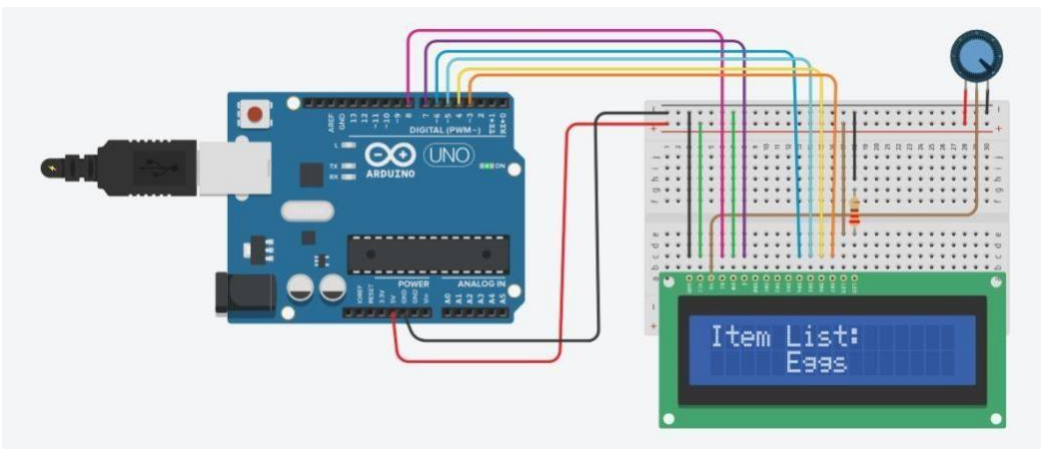
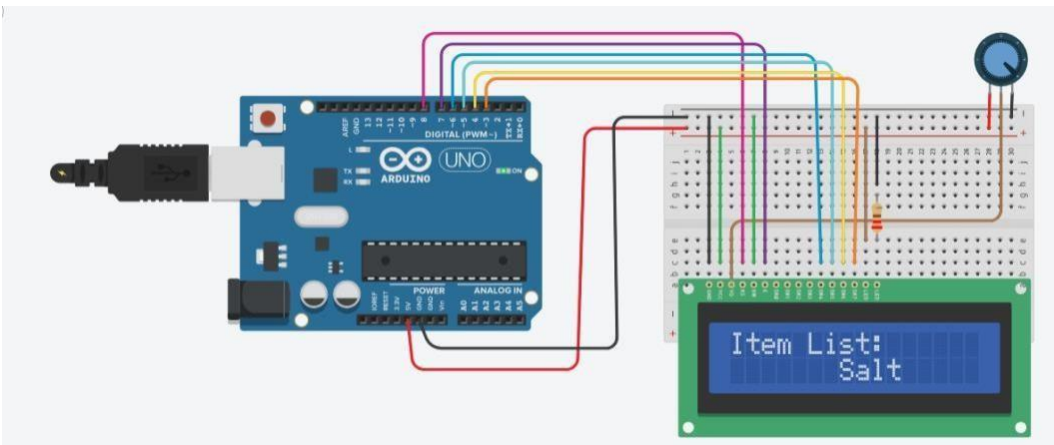
Class: TYIT(NEP) Sem: V
Course Name:Internet Of Things
Practical Number:-2b

Roll No.: TYIT16

Date:
Page no:



Output:



Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class: TYIT

Sem: V

Roll No.: TYIT16

Date :21/07/2026

Course Name:IOT

Page no:

Practical Number:-2

AIM 2(c): Interfacing 16x2 LCD and pushbutton with Arduino.

Code:

```
#include <LiquidCrystal.h>
```

```
LiquidCrystal lcd(8, 7, 6, 5, 4, 3);
```

```
const int buttonPin = 2;
```

```
bool lcdState = false;
```

```
bool lastButtonState = HIGH;
```

```
void setup() {  
  lcd.begin(16, 2);  
  pinMode(buttonPin, INPUT_PULLUP);  
  lcd.noDisplay();    // LCD starts OFF  
}
```

```
void loop() {  
  bool currentButtonState = digitalRead(buttonPin);  
  
  if (lastButtonState == HIGH && currentButtonState == LOW) {  
    lcdState = !lcdState;    // Toggle LCD state  
  
    if (lcdState) {  
      lcd.display();  
      lcd.clear();  
      lcd.setCursor(0, 0);  
      lcd.print("LCD ON!");  
      lcd.setCursor(0, 1);  
      lcd.print("Hello TYIT16");  
    } else {  
      lcd.noDisplay();  
    }  
  
    delay(200); // Debounce  
  }  
  
  lastButtonState = currentButtonState;  
}
```

Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class: TYIT

Sem: V

Roll No.: TYIT16

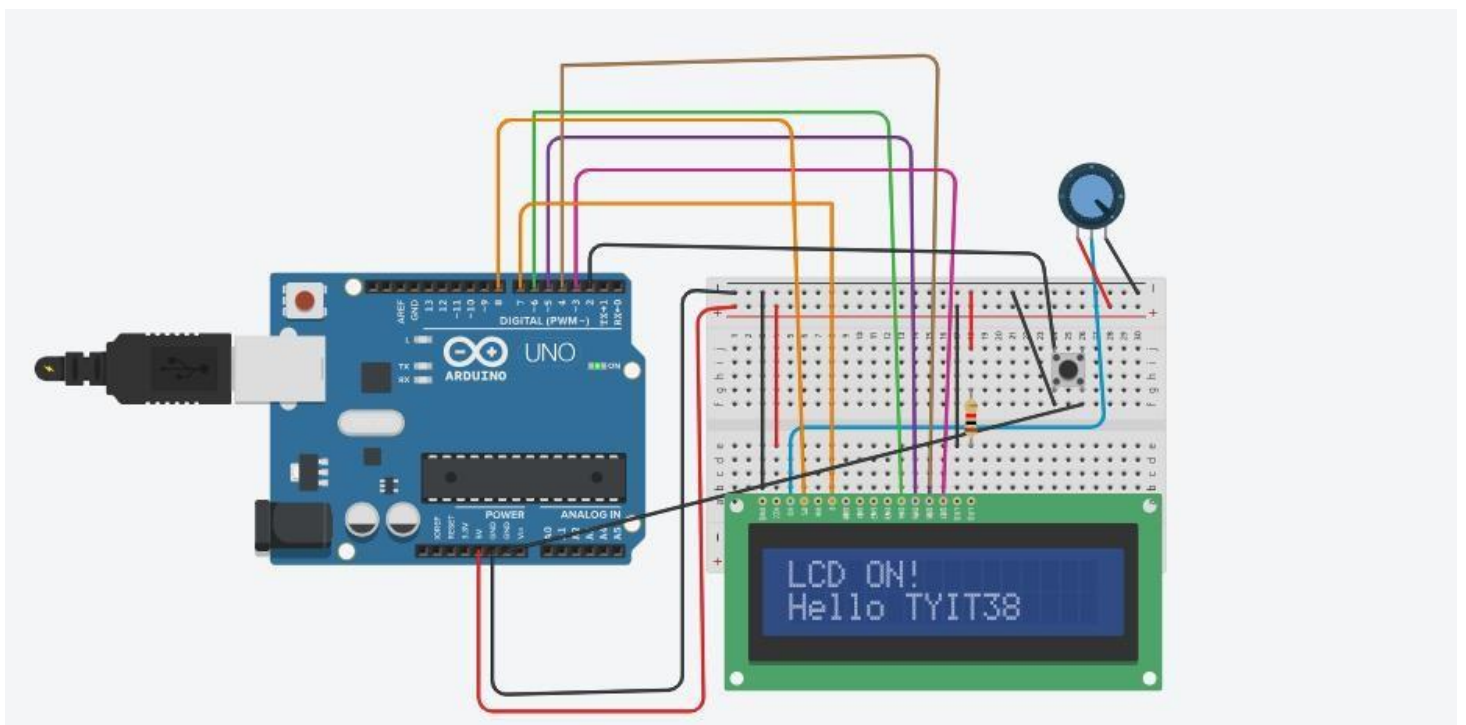
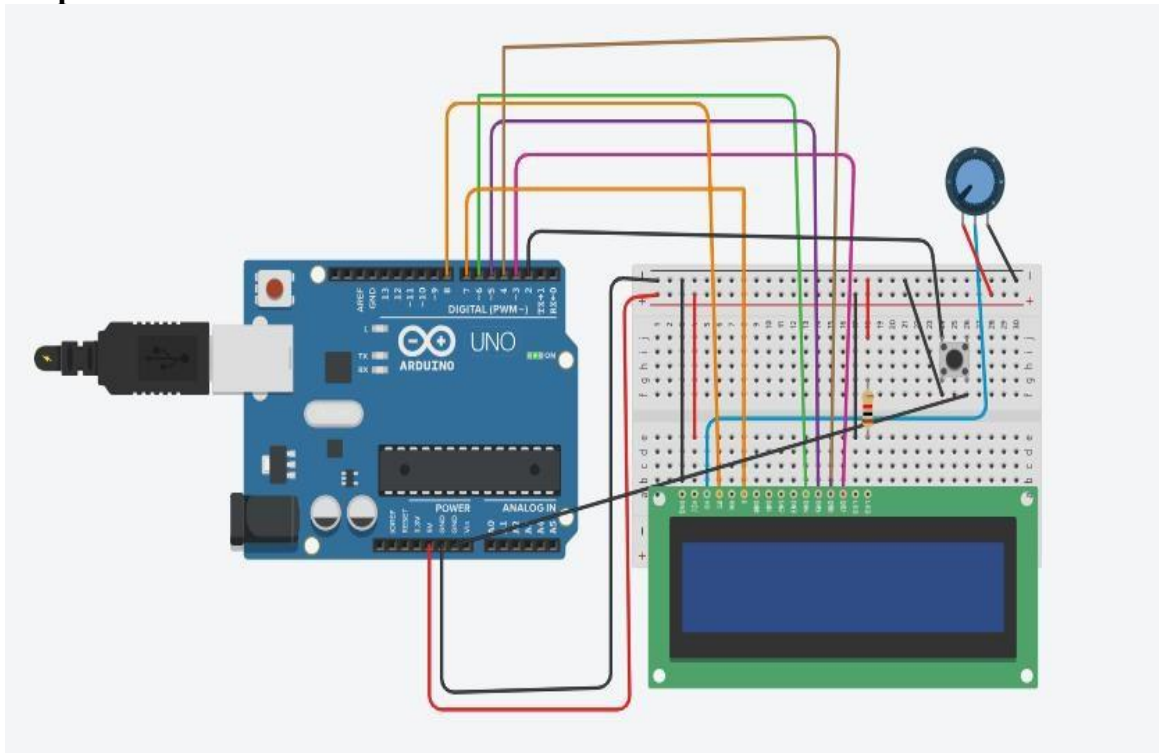
Date :21/07/2026

Course Name:IOT

Page no:

Practical Number:-2

Output:



Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class :- TYIT

Sem :- V

Roll No. :- TYIT16

Date :- 01/08/2026

Course Name :- Internet Of Things

Page no :-

Practical Number :- 2D

Aim: Design and implement a small doorbell using a push button in tinkercad when the button is pressed activate a buzzer, light an LED, and display "Doorbell Pressed" on the LCD to display to simulate a smartphone notification

Steps:

1. Open Tinkercad and create a new blank Circuit.
2. Drag an Arduino Uno R3, a breadboard, a 16×2 LCD (parallel interface), a 10 kΩ potentiometer, a 220 Ω resistor, a LED, a push button, and a piezo buzzer onto the workspace.
3. Place the 16×2 LCD on the breadboard so that its 16 pins are inserted into separate rows.
4. Connect the Arduino power pins:
 - Arduino 5V → Breadboard positive (+) rail
 - Arduino GND → Breadboard negative (−) rail
5. Power the LCD:
 - LCD Pin 1 (VSS) → GND rail
 - LCD Pin 2 (VDD) → +5V rail
 - LCD Pin 15 (LED+) → +5V through a 220 Ω resistor
 - LCD Pin 16 (LED−) → GND rail
6. Connect the contrast control using the 10 kΩ potentiometer:
 - One outer terminal → +5V rail
 - Other outer terminal → GND rail
 - Middle terminal (wiper) → LCD Pin 3 (VO)
7. Connect the LCD control pins to the Arduino:
 - LCD Pin 4 (RS) → Arduino Digital Pin 12
 - LCD Pin 5 (RW) → GND
 - LCD Pin 6 (E) → Arduino Digital Pin 11
8. Connect the LCD data pins in 4-bit mode:
 - LCD Pin 11 (D4) → Arduino Digital Pin 5
 - LCD Pin 12 (D5) → Arduino Digital Pin 4
 - LCD Pin 13 (D6) → Arduino Digital Pin 3
 - LCD Pin 14 (D7) → Arduino Digital Pin 2
 - Leave LCD Pins 7–10 (D0–D3) unconnected.
9. Connect the LED:
 - Arduino digital output pin → 330 Ω resistor
 - Resistor → LED anode (+)
 - LED cathode (−) → GND
10. Connect the push button:
 - Place the push button across the center gap of the breadboard.
 - One side → Arduino digital input pin
 - Other side → GND
 - Configure the input as INPUT_PULLUP in the Arduino program if using this connection.
11. Connect the piezo buzzer:
 - Buzzer positive (+) → Arduino digital output pin
 - Buzzer negative (−) → GND
12. Write and upload the Arduino program using the LiquidCrystal library. For the LCD connections above, use: LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
13. Start the simulation in Tinkercad.

Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class :- TYIT

Sem :- V

Roll No. :- TYIT16

Date :- 01/08/2026

Course Name :- Internet Of Things

Page no :-

Practical Number :- 2D

14. Rotate the 10 kΩ potentiometer until the LCD characters become visible.
15. Test the circuit by pressing the push button and checking the LED and buzzer response programmed in the Arduino sketch.
16. . Verify that the LCD displays the required text or values and that all components operate correctly.

Code:-

```
#include <LiquidCrystal.h>
```

```
// LCD Pins: RS, E, D4, D5, D6, D7
LiquidCrystal lcd(8, 7, 6, 5, 4, 3);
```

```
// Component Pins
const int buttonPin = 2;
const int ledPin = 9;
const int buzzerPin = 10;
```

```
void setup() {
  pinMode(buttonPin, INPUT_PULLUP);
  pinMode(ledPin, OUTPUT);
  pinMode(buzzerPin, OUTPUT);
```

```
  digitalWrite(ledPin, LOW);
  noTone(buzzerPin);
```

```
  lcd.begin(16, 2);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Smart Doorbell");
  lcd.setCursor(0, 1);
  lcd.print("Waiting...");
}
```

```
void loop() {
```

```
  // Button pressed (INPUT_PULLUP -> LOW)
  if (digitalRead(buttonPin) == LOW) {
```

```
    // Turn ON LED
    digitalWrite(ledPin, HIGH);
```

```
    // Turn ON buzzer
    tone(buzzerPin, 1000);
```

```
    // Display notification
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Doorbell TYIT16");
    lcd.setCursor(0, 1);
```

Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class :- TYIT

Sem :- V

Roll No. :- TYIT16

Date :- 01/08/2026

Course Name :- Internet Of Things

Page no :-

Practical Number :- 2D

```
lcd.print("Pressed");

// Keep everything ON for 3 seconds
delay(3000);

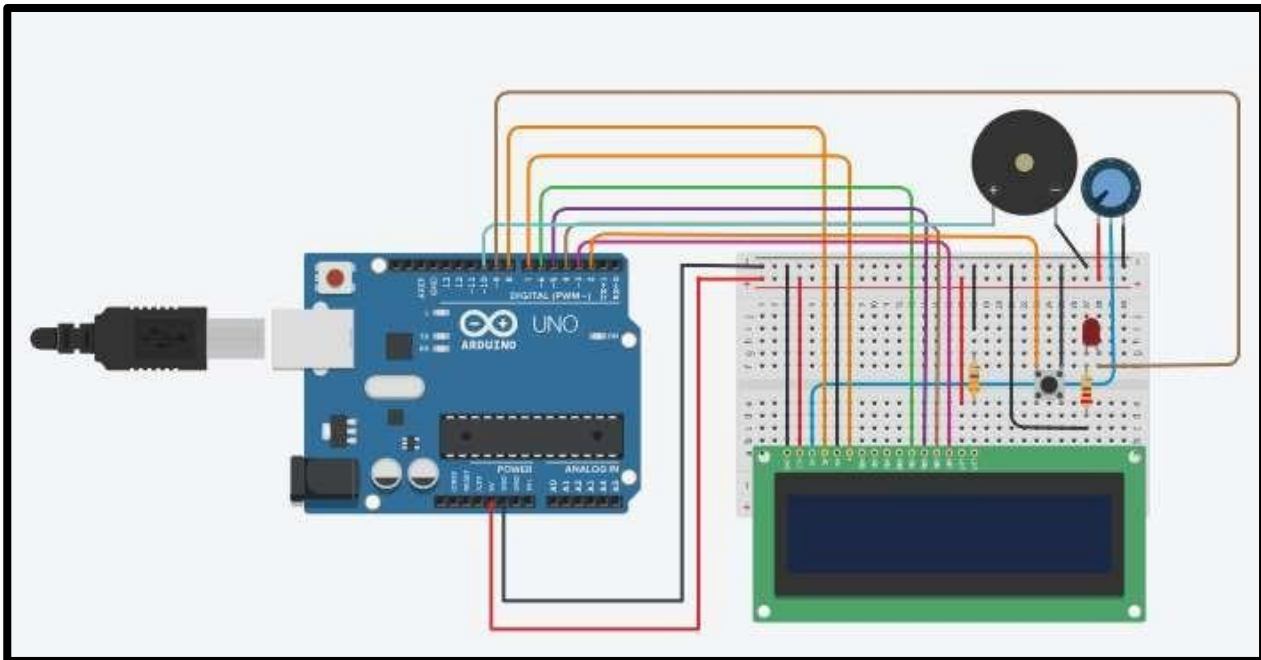
// Turn OFF buzzer and LED
noTone(buzzerPin);
digitalWrite(ledPin, LOW);

// Return to waiting screen
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Smart Doorbell");
lcd.setCursor(0, 1);
lcd.print("Waiting...");

// Wait until button is released
while (digitalRead(buttonPin) == LOW);

delay(200); // Debounce
}
}
```

Circuit Diagram:



Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class :- TYIT

Sem :- V

Roll No. :- TYIT16

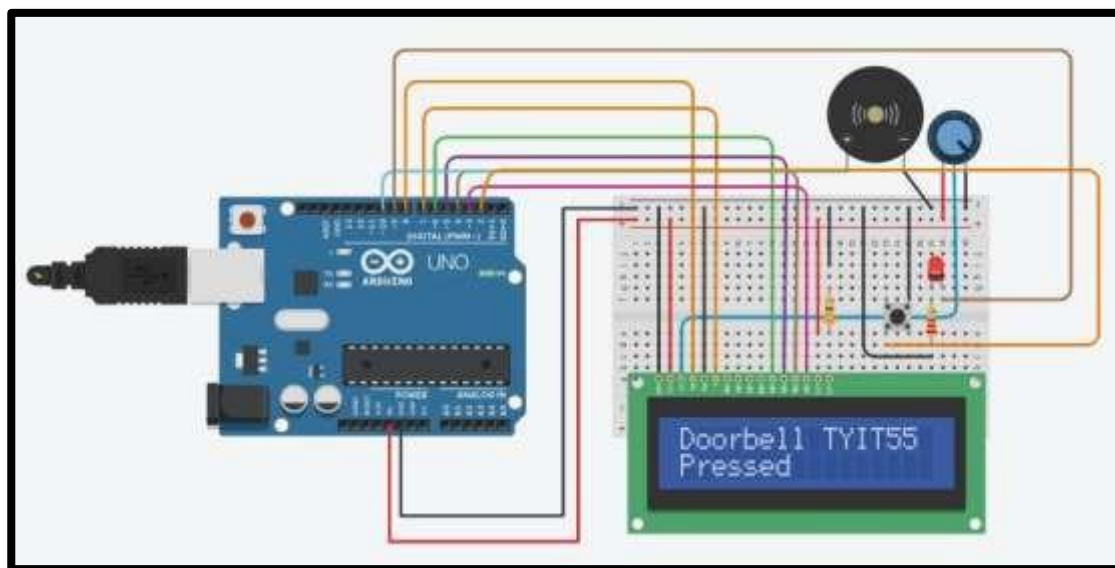
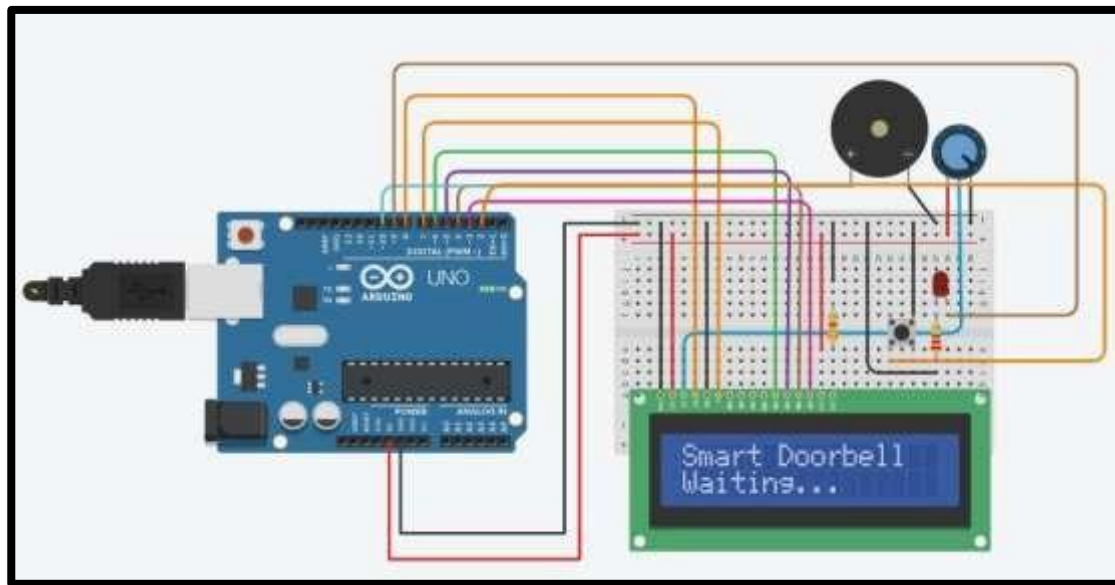
Date :- 01/08/2026

Course Name :- Internet Of Things

Page no :-

Practical Number :- 2D

Output:-



Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class :- TYIT

Sem :- V

Roll No. :- TYIT16

Date :- 08/08/2026

Course Name :- Internet Of Things

Page no :-

Practical Number :- 3A

Aim: Design and implement a parking assistance system using an ultrasonic sensor. Measure the distance to an obstacle and activate a buzzer when the object is too close.

Steps:

Open Tinkercad and create a new blank Circuit.

1. Drag an Arduino Uno R3, a breadboard, an HC-SR04 ultrasonic sensor, and a piezo
2. buzzer onto the workspace.
3. Place the HC-SR04 ultrasonic sensor on the breadboard/workspace so that its four pins are accessible.
4. Connect the Arduino power pins:
 - Arduino 5V → Breadboard positive (+) rail
 - Arduino GND → Breadboard negative (-) rail
5. Connect the HC-SR04 power pins:
 - HC-SR04 VCC → +5V rail
 - HC-SR04 GND → GND rail
6. Connect the HC-SR04 control pins to the Arduino:
 - HC-SR04 TRIG → Arduino Digital Pin 7
 - HC-SR04 ECHO → Arduino Digital Pin 6
7. Place the piezo buzzer on the breadboard and connect it:
 - Buzzer positive (+) → Arduino Digital Pin 8
 - Buzzer negative (-) → GND rail
8. Verify all connections:
 - 5V supplies power to the HC-SR04.
 - GND is shared by the Arduino, sensor, and buzzer.
 - D7 is used for the ultrasonic sensor TRIG pin.
 - D6 is used for the ultrasonic sensor ECHO pin.
 - D8 controls the buzzer.
9. Write and upload the Arduino program to measure the distance using the HC-SR04 and
10. control the buzzer.

Code:-

```
int buzPin=7;
int trigPin=8;
int echoPin = 4;
float speed = 0.0347;
float dist;
float pingTime;
int buzNear=20;
int buzHigh=50;
int buzMid=130;
int buzFar=600;
int delayFar = 260;
```

```
void setup()
{
  Serial.begin(9600);
```

Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class :- TYIT

Sem :- V

Roll No. :- TYIT16

Date :- 08/08/2026

Course Name :- Internet Of Things

Page no :-

Practical Number :- 3A

```
pinMode(buzPin, OUTPUT);
pinMode(trigPin, OUTPUT);
pinMode(echoPin, INPUT);
}

void loop()
{
  digitalWrite(trigPin, LOW);
  delayMicroseconds(20);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  pingTime = pulseIn(echoPin, HIGH);
  dist = (speed*pingTime*0.5);
  Serial.print("Practical performed by Namrata TYIT16:");
  Serial.println(dist);

  if(dist<=10.0)
  {
    digitalWrite(buzPin, HIGH);
    delay(20);
    digitalWrite(buzPin, LOW);
    delay(20);
  }
  else if(dist<=3.0 && dist>10.0)
  {
    digitalWrite(buzPin, HIGH);
    delay(buzHigh);
    digitalWrite(buzPin, LOW);
    delay(buzHigh);
  }
  else if((dist>30.0) && (dist<60.0))
  {
    digitalWrite(buzPin, HIGH);
    delay(buzMid);
    digitalWrite(buzPin, LOW);
    delay(buzMid);
  }
  else if ((dist>=60.0 && dist<120.0))
  {
    digitalWrite(buzPin, HIGH);
    delay(buzFar);
    digitalWrite(buzPin, LOW);
    delay(delayFar);
  }
}
```

Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class :- TYIT

Sem :- V

Roll No. :- TYIT16

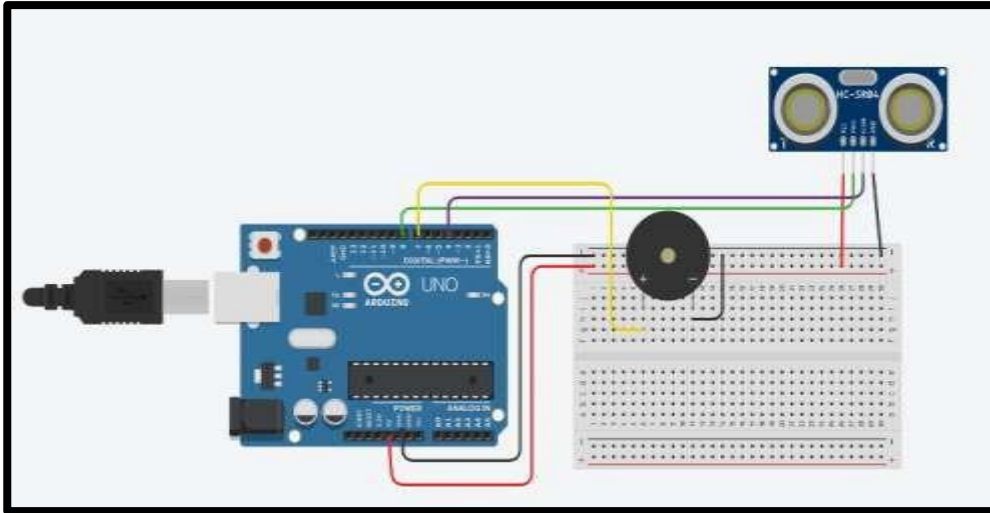
Date :- 08/08/2026

Course Name :- Internet Of Things

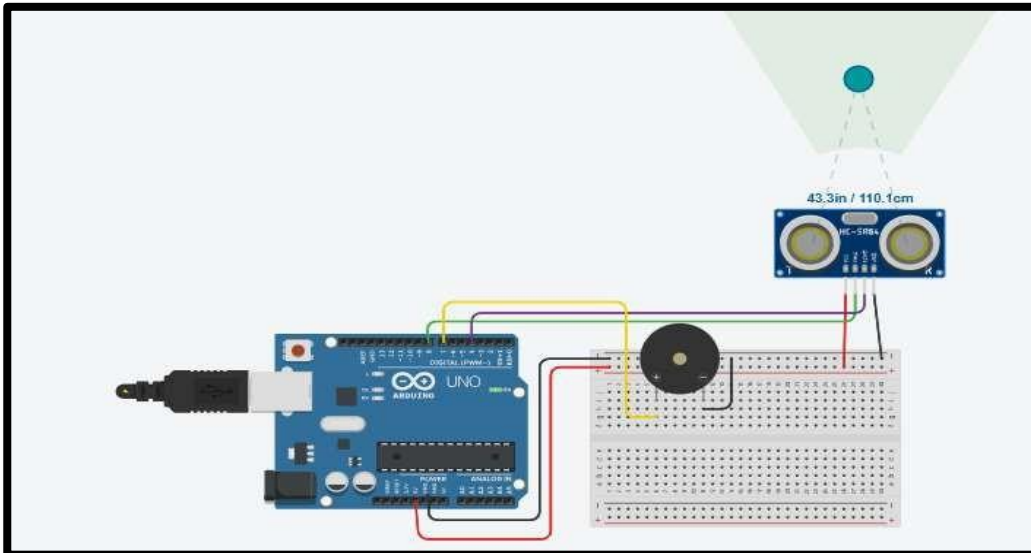
Page no :-

Practical Number :- 3A

Circuit Diagram:-



Output:-



Serial Monitor

```
Practical performed by Namrata TYIT55:117.86
Practical performed by Namrata TYIT55:118.05
Practical performed by Namrata TYIT55:117.86
Practical performed by Namrata TYIT55:118.01
Practical performed by Namrata TYIT55:118.05
Practical performed by Namrata TYIT55:118.05
Practical performed by Namrata TYIT55:118.05
Practical performed by Namrata TYIT55:118.05
Practical performed by Namrata TYIT55:118.01
Practical performed by Namrata TYIT55:117.86
Practical performed by Namrata TYIT55:118.05
Practical performed by Namrata TYIT55:117.86
Practical performed by Namrata TYIT55:118.05
```

Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class: TYIT

Sem: V

Roll No.: TYIT16

Date :19/08/26

Course Name:NGT

Page no:

Practical Number:-IOT

AIM: Design and implement a temperature monitoring system using the TMP36 sensor. Display the temperature on the Serial Monitor and glow an LED whenever the temperature exceeds a predefined threshold.

STEPS:

1. Connect the TMP36 temperature sensor to A0.
2. Connect the Red LED to pin 10.
3. Connect the Blue LED to pin 12.
4. Start the Arduino and read the temperature from the sensor.
5. Convert the sensor value into temperature in °C.
6. Show the temperature on the Serial Monitor.
7. If temperature is above 50°C, turn Red LED ON.
8. If temperature is below 0°C, turn Blue LED ON.
9. If temperature is between 0°C and 50°C, keep both LEDs OFF.
10. Repeat the process every 500 milliseconds.

CODE:

```
// Temperature-Controlled LED System
const int tempPin = A0; const
int redLED = 10;
const int blueLED = 12;

void setup()
{
  pinMode(redLED, OUTPUT);
  pinMode(blueLED, OUTPUT);

  Serial.begin(9600);
}

void loop()
{
  // Read the sensor value (0-1023) int
  sensorValue = analogRead(tempPin);

  // Convert the reading to voltage float
  voltage = sensorValue * (5.0 / 1023.0);

  // Convert the voltage to temperature (TMP36)
  float temperature = (voltage - 0.5) * 100.0;

  // Display the temperature in the Serial Monitor
  Serial.println("Practical Performed by Bhumi ");
  Serial.print("Current Temperature is: ");
  Serial.print(temperature);
  Serial.println(" °C");
```

Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class: TYIT

Sem: V

Roll No.: TYIT16

Date :19/08/26

Course Name:IOT

Page no:

Practical Number:-

Serial.println();

// Temperature above 50°C → Red LED ON

if (temperature > 50)

{

digitalWrite(redLED, HIGH);

digitalWrite(blueLED, LOW);

}

// Temperature below 0°C → Blue LED ON

else if (temperature < 0)

{

digitalWrite(blueLED, HIGH);

digitalWrite(redLED, LOW);

}

// Temperature between 0°C and 50°C → Both LEDs OFF

else

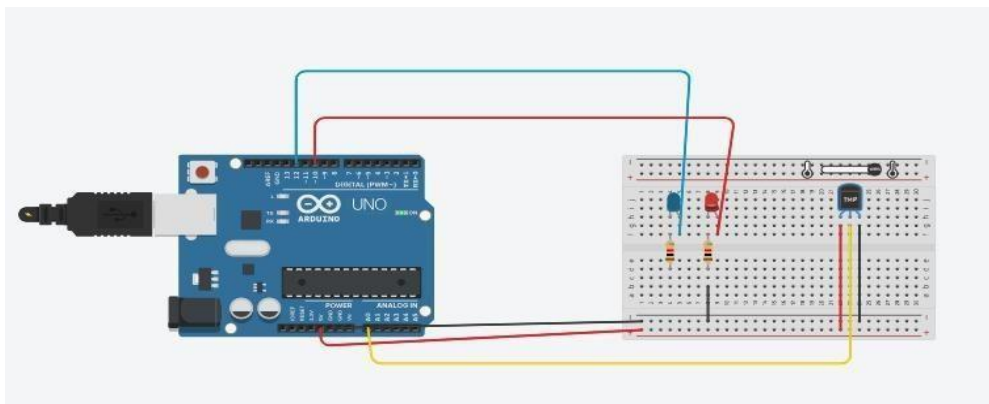
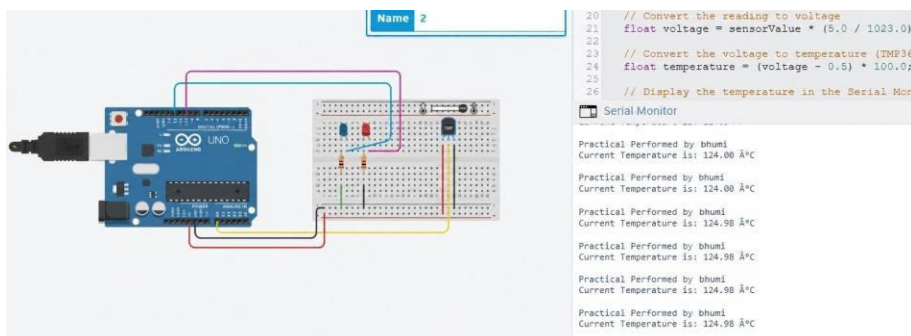
{

digitalWrite(redLED, LOW);

digitalWrite(blueLED, LOW);

}

delay(500);



Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class: TYIT

Sem:V

Roll No.: TYIT16

Date : 02/09/2026

Course Name:IOT

Page no:

Practical Number:-

AIM: Design and implement a gas leakage monitoring system using an MQ-2 gas sensor. If gas concentration exceeds a threshold, switch ON an LED, activate a buzzer, and display a warning message on the Serial Monitor.

STEPS:

1. Connect the MQ-2 gas/smoke sensor to analog pin A3.
2. Connect 5 LEDs to pins 2, 3, 4, 5, and 6.
3. Connect the buzzer to pin 12.
4. Start the Arduino and read the value from the MQ-2 sensor.
5. Display the sensor value on the Serial Monitor.
6. If the value is below 970, all LEDs and the buzzer remain OFF.
7. If the value is 970–978, 1 LED turns ON.
8. If the value is 979–986, 2 LEDs turn ON.
9. If the value is 987–994, 3 LEDs turn ON.
10. If the value is 995–1003, 4 LEDs turn ON.
11. If the value is 1004 or above, all 5 LEDs and the buzzer turn ON.
12. The system repeats the process every 200 milliseconds.

Code:

```
// MQ-2 Smoke/Gas Detection System  
// Tinkercad sensor range: approximately 970 to 1013
```

```
const int gasSensor = A3;
```

```
const int led1 = 2;  
const int led2 = 3;  
const int led3 = 4;  
const int led4 = 5;  
const int led5 = 6;
```

```
const int buzzer = 12;
```

```
int gasValue;
```

```
void setup()  
{  
  pinMode(led1, OUTPUT);  
  pinMode(led2, OUTPUT);  
  pinMode(led3, OUTPUT);  
  pinMode(led4, OUTPUT);  
  pinMode(led5, OUTPUT);  
  
  pinMode(buzzer, OUTPUT);
```

Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class: TYIT

Sem:V

Roll No.: TYIT16

Date : 02/09/2026

Course Name:IOT

Page no:

Practical Number:-

```
Serial.begin(9600);
}

void loop()
{
  // Read MQ-2 sensor
  gasValue = analogRead(gasSensor);

  Serial.print("Smoke Detected at radius: ");
  Serial.println(gasValue);
  Serial.println("Practical Performed by Bhumi Pandey");
  Serial.println();

  // Turn OFF all LEDs and buzzer
  digitalWrite(led1, LOW);
  digitalWrite(led2, LOW);
  digitalWrite(led3, LOW);
  digitalWrite(led4, LOW);
  digitalWrite(led5, LOW);
  digitalWrite(buzzer, LOW);

  // Very low smoke
  if (gasValue < 970)
  {
    // All LEDs OFF
  }

  // Stage 1
  else if (gasValue < 979)
  {
    digitalWrite(led1, HIGH);
  }

  // Stage 2
  else if (gasValue < 987)
  {
    digitalWrite(led1, HIGH);
    digitalWrite(led2, HIGH);
  }

  // Stage 3
  else if (gasValue < 995)
  {
    digitalWrite(led1, HIGH);
```

Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class: TYIT

Sem:V

Roll No.: TYIT16

Date : 02/09/2026

Course Name:IOT

Page no:

Practical Number:-

```
digitalWrite(led2, HIGH);
digitalWrite(led3, HIGH);
}

// Stage 4
else if (gasValue < 1004)
{
    digitalWrite(led1, HIGH);
    digitalWrite(led2, HIGH);
    digitalWrite(led3, HIGH);
    digitalWrite(led4, HIGH);
}

// Maximum smoke/gas concentration
else
{
    digitalWrite(led1, HIGH);
    digitalWrite(led2, HIGH);
    digitalWrite(led3, HIGH);
    digitalWrite(led4, HIGH);
    digitalWrite(led5, HIGH);

    digitalWrite(buzzer, HIGH);
}

delay(200);
}
```

Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class: TYIT

Sem: V

Roll No.: TYIT16

Date : 02/09/2026

Course Name: IOT

Page no:

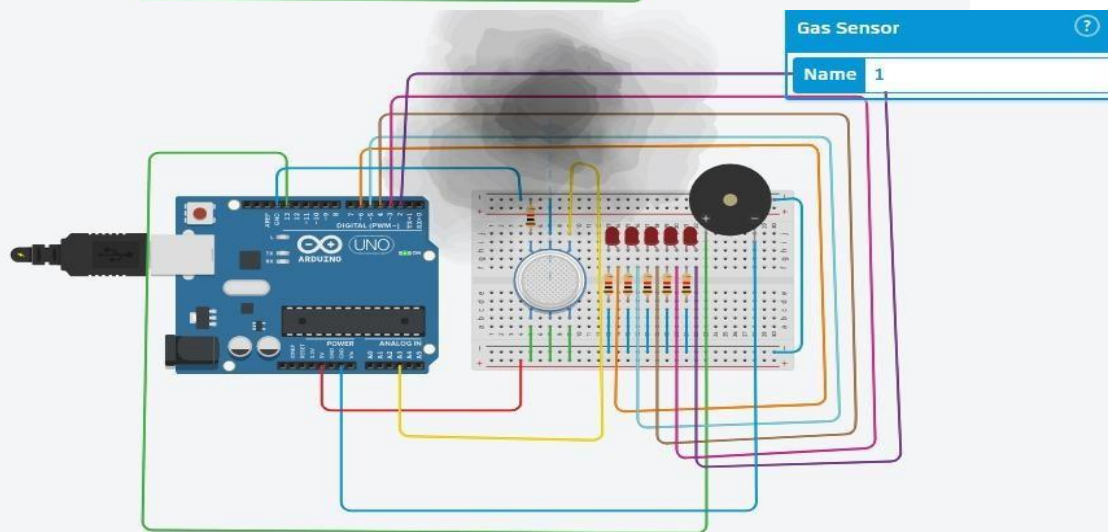
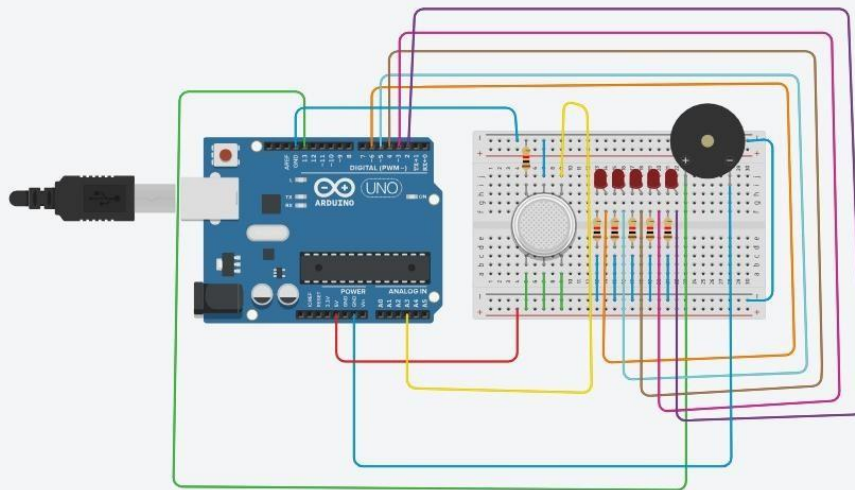
Practical Number:-

Output:

Serial Monitor

Smoke Detected at radius: 101
Practical Performed by Bhumi Pandey

Smoke Detected at radius: 101
Practical Performed by Bhumi Pandey



Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class: TYIT(NEP)

Sem: V

Roll No.: TYIT16

Date: 02/09/2026

Course Name: IoT

Page no:

Practical Number: -05

Practical aim : Starting Raspbian OS, Familiarizing with Raspberry Pi Components and interface, Connecting to ethernet, Monitor, USB

1. Introduction of Raspberry Pi.

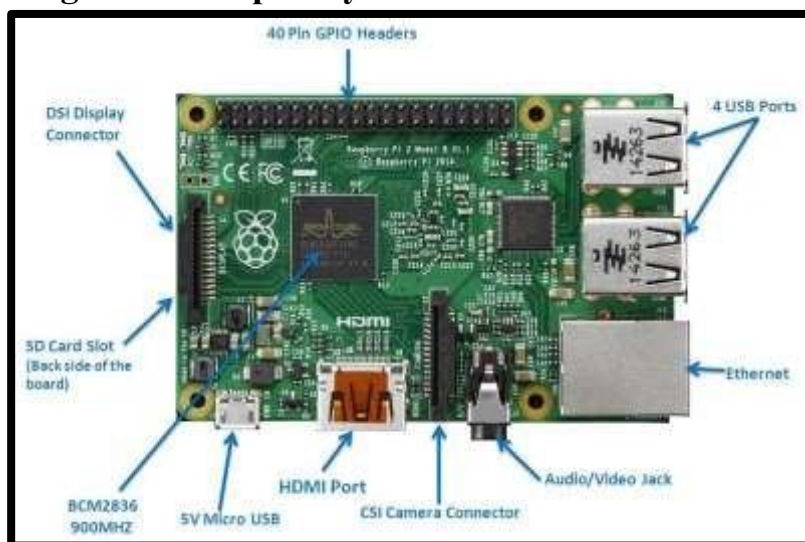
Raspberry Pi is a small, low-cost single-board computer used for programming, electronics, IoT, and automation applications. It can be connected to peripherals such as a keyboard, mouse, monitor, and USB devices. It also provides GPIO pins for connecting electronic components and sensors.

In this practical, the Raspberry Pi is connected with a keyboard, mouse, monitor, and power supply to set up and operate it as a basic computer.

2. Various components of Raspberry pi

- **Processor (CPU):** Performs calculations and executes instructions.
- **RAM:** Stores data and programs temporarily while the Raspberry Pi is running.
- **GPIO Pins:** Used to connect sensors, LEDs, motors, and other electronic components.
- **USB Ports:** Used to connect devices such as a keyboard, mouse, pen drive, or other USB devices.
- **HDMI Port:** Connects the Raspberry Pi to a monitor or TV for display.
- **Power Port:** Supplies power to the Raspberry Pi, usually through a USB-C or micro-USB connector depending on the model.
- **MicroSD Card Slot:** Used to store the operating system, programs, and user data.
- **Ethernet/Wi-Fi:** Provides network and internet connectivity.
- **Audio/Video Interface:** Allows audio and video output through supported display connections.

3. Diagram of Raspberry Pi



Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class: TYIT(NEP)

Sem: V

Roll No.: TYIT16

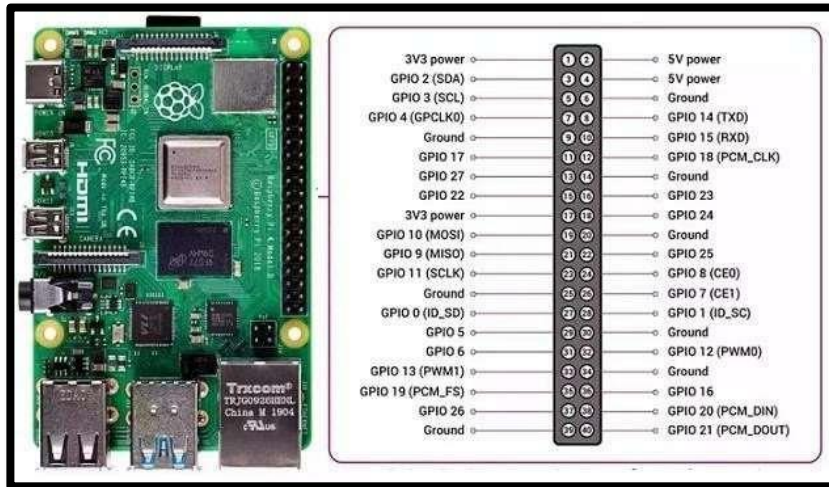
Date: 02/09/2026

Course Name: IoT

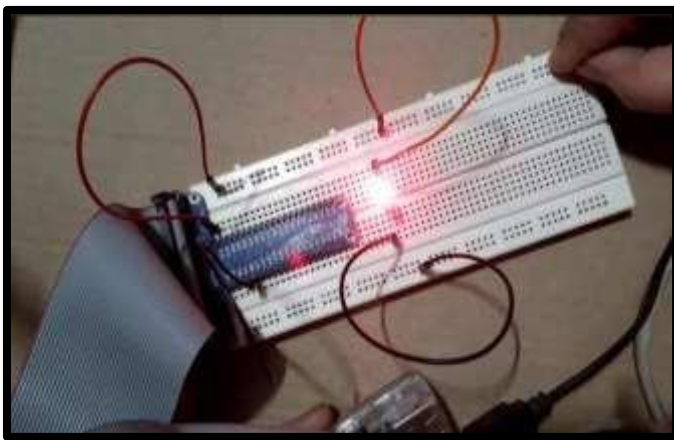
Page no:

Practical Number: -05

4. Pin diagram of Raspberry Pi



5. Output: -



Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class: TYIT

Sem: V

Roll No.: TYIT16

Date: 09/09/2026

Course Name: IOT

Page no:

Practical Number: - 06

AIM: Interface Raspberry Pi with LED Module to demonstrate LED pattern.

Setup:

- 1] Connect Raspberry Pi with Keyboard, Mouse, Monitor and Ethernet.
- 2] Take eight jumper wires [female to female] and connects them to LED and GPIO pins. 3] Take one jumper wire and connect it to the ground.

Code:

```
import RPi.GPIO as GPIO
import time

# Use physical pin numbering
GPIO.setmode(GPIO.BOARD)

# LED connections using physical pin numbers leds
= [37, 35, 33, 31, 29, 32, 36, 38]

# Set LED pins as output for led
in leds:
    GPIO.setup(led, GPIO.OUT)
    GPIO.output(led, GPIO.LOW)

try:    while True:
        # LED pattern: 1 → 2 → 3 → 4
        for led in leds:
            GPIO.output(led, GPIO.HIGH)
            time.sleep(0.5)
            GPIO.output(led, GPIO.LOW)

        # LED pattern: 4 → 3 → 2 → 1
        for led in reversed(leds):
            GPIO.output(led, GPIO.HIGH)
            time.sleep(0.5)
            GPIO.output(led, GPIO.LOW)

except KeyboardInterrupt:    print("Program
stopped")

finally:
    GPIO.cleanup()
```

Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class: TYIT

Sem: V

Roll No.: TYIT16

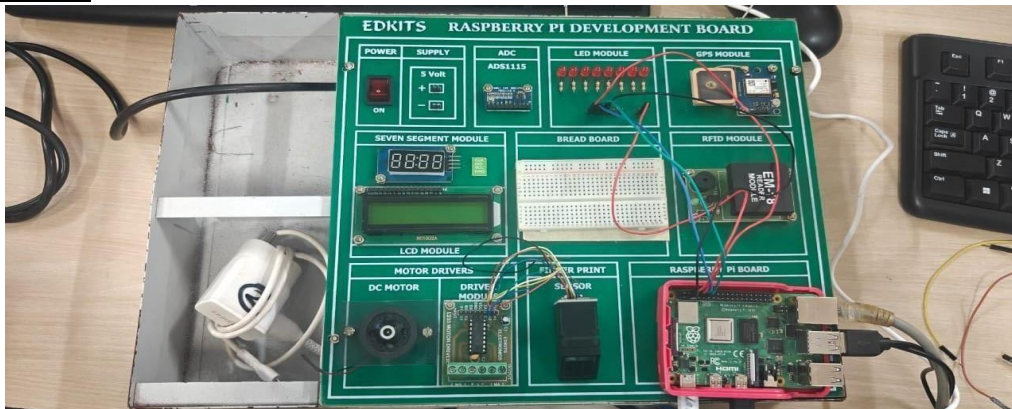
Date: 09/09/2026

Course Name: IOT

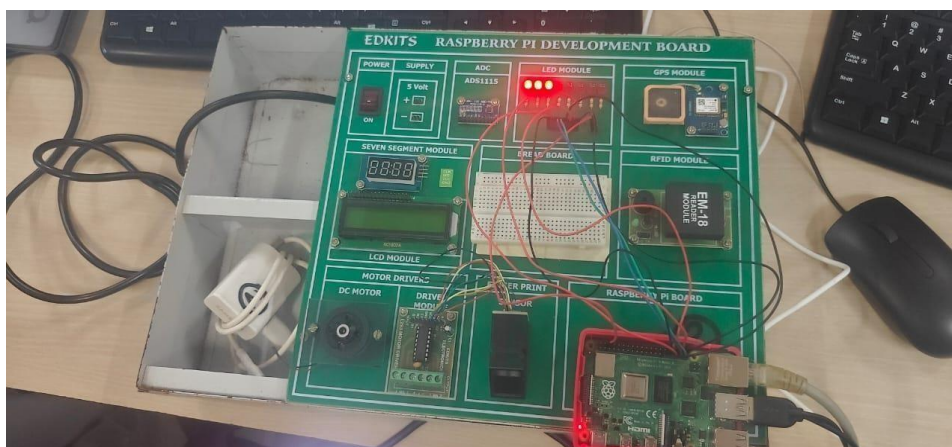
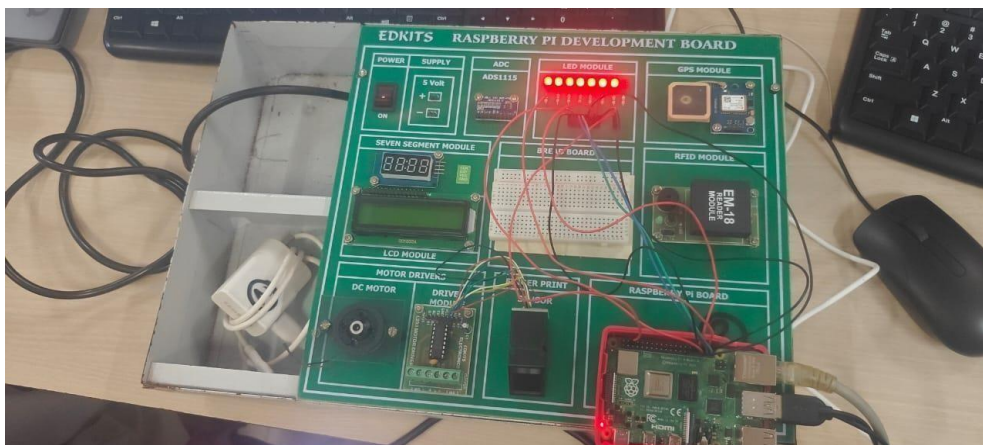
Page no:

Practical Number: - 06

Circuit:



Output:



Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST

PRACTICAL JOURNAL

Class: TYIT(NEP)

Sem: V

Roll No.: TYIT16

Date: 09/09/2026

Course Name: Internet of Things

Page no:

Practical Number:- 7

AIM: DISPLAYING TIME OVER A 4-DIGIT SEVEN SEGMENT DISPLAY USING RASPBERRY PI.

○ STEPS:

- 1) Connect the **Raspberry Pi** to a **keyboard, mouse, and Ethernet cable**.
- 2) Connect the **monitor** to the Raspberry Pi using an **HDMI cable**.
- 3) Connect the **GND pin** of the TM1637 display module to a **GND pin** of the Raspberry Pi using a **female-to-female jumper wire**.
- 4) Connect the **VCC/Power pin** of the TM1637 display module to the appropriate **power pin** of the Raspberry Pi using a **female-to-female jumper wire**.
- 5) Connect the **DIO pin** of the TM1637 module to **physical pin 18 (GPIO 24)** of the Raspberry Pi.
- 6) Connect the **CLK pin** of the TM1637 module to **physical pin 16 (GPIO 23)** of the Raspberry Pi.
- 7) Connect the Raspberry Pi to a **power supply using a USB power cable**.
- 8) Open **Thonny Python IDE** on the Raspberry Pi.
- 9) Type or paste the required Python program into Thonny.
- 10) Make sure that **tm1637.py** and **pract7.py** are saved in the **same folder/location**.
- 11) Save the program as **pract7.py**.
- 12) Click **Run** in Thonny to execute the program.
- 13) Observe the output on the **TM1637 display module**.

○ CODE:

• pract7.py

```
import sys
import time
import datetime
import RP1.GPIO as GPIO
import tm1637
```

```
#CLK GPIO23 (Pin 16)
```

```
#D10 GP1024 (Pin 18)
```

```
Display = tm1637.TM1637(23, 24, tm1637.BRIGHT_TYPICAL)
```

```
Display.Clear()
```

```
Display.SetBrightness(1)
```

```
while(True):
```

```
    now = datetime.datetime.now()
    hour
```

```
    = now.hour
```


BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST PRACTICAL JOURNAL

Class: TYIT(NEP)

Sem: V

Roll No.: TYIT16

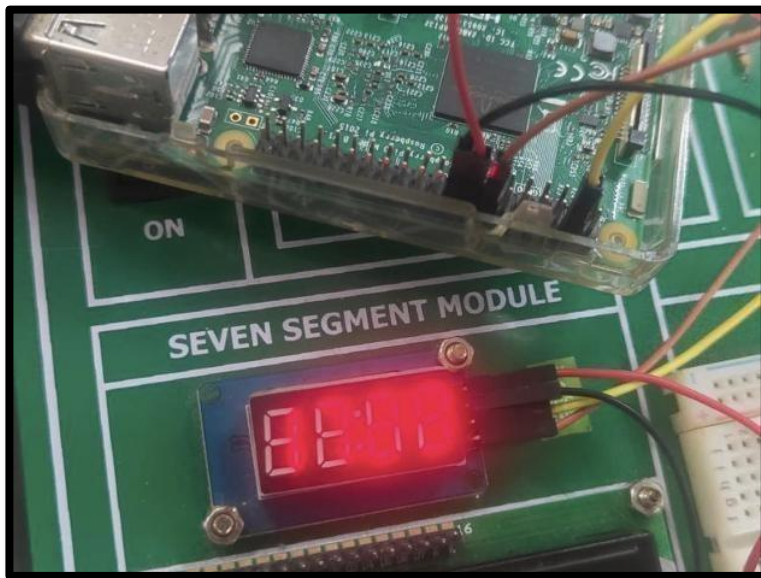
Date: 09/09/2026

Course Name: Internet of Things

Page no:

Practical Number:- 7 minute = now.minute second = now.second
currenttime = [int(hour/10), hour % 10, int(minute/10), minute % 10]
Display. Show(currenttime)
Display. ShowDoublepoint(second % 2) time.sleep(1)

○ CIRCUIT DIAGRAM:



○ OUTPUT:

